Predicting Restaurant Inspection Risk

Using Historical Inspection Data - Los Angeles County

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Why Restaurant Inspection Grades Matter

Publicly Posted

Grades are displayed at every entrance. Customers see them before they decide to eat.

Revenue Impact

A downgrade from A to B directly affects customer traffic, revenue, and long-term viability.

Beyond Compliance

Inspection outcomes are operationally meaningful - not just regulatory checkboxes.

The Core Question: Can historical inspection data predict future non-A outcomes?

Dataset Overview - LA County Environmental Health

106,694
Total Inspections
(A, B, C grades)

4,861
Non-A Outcomes
(4.6% of total)

63,782
Rows for Modeling
(≥1 prior inspection)

2023–2025
Date Range
Covered

⚠ Class Imbalance Challenge

Only 4.6% of inspections result in a non-A grade, creating a highly imbalanced dataset. Standard accuracy metrics are misleading - a model that always predicts 'A' would score 95.4%. Evaluation must rely on ROC-AUC, Precision, Recall, and F1.

Feature Engineering - Building the Historical Signal

prior_avg_score

Rolling average score from all previous inspections at this facility

prior_nonA_rate

Historical rate of non-A grades at this specific facility

prior_inspection_count

Total number of prior inspections on record for the facility

days_since_last_inspection

Calendar days elapsed since the facility's most recent inspection

Only rows with at least 1 prior inspection were used - first-time facilities have no historical signal.

Model Comparison — Three Classifiers Tested

Model	ROC-AUC	Precision	Recall	F1
Gradient Boosting ⭐	0.785	0.491	0.028	0.052
Random Forest	0.755	0.475	0.037	0.069
Logistic Regression	0.741	0.183	0.530	0.272

Key Takeaways

- Gradient Boosting achieved the highest ROC-AUC (0.785), indicating the strongest overall discrimination.
- Logistic Regression had the highest recall (0.530) - useful if catching non-A cases is the priority, at the cost of many false alarms.
- All models struggle with the rare-event nature of non-A grades. Threshold tuning is a logical next step.

Gradient Boosting — Confusion Matrix Analysis

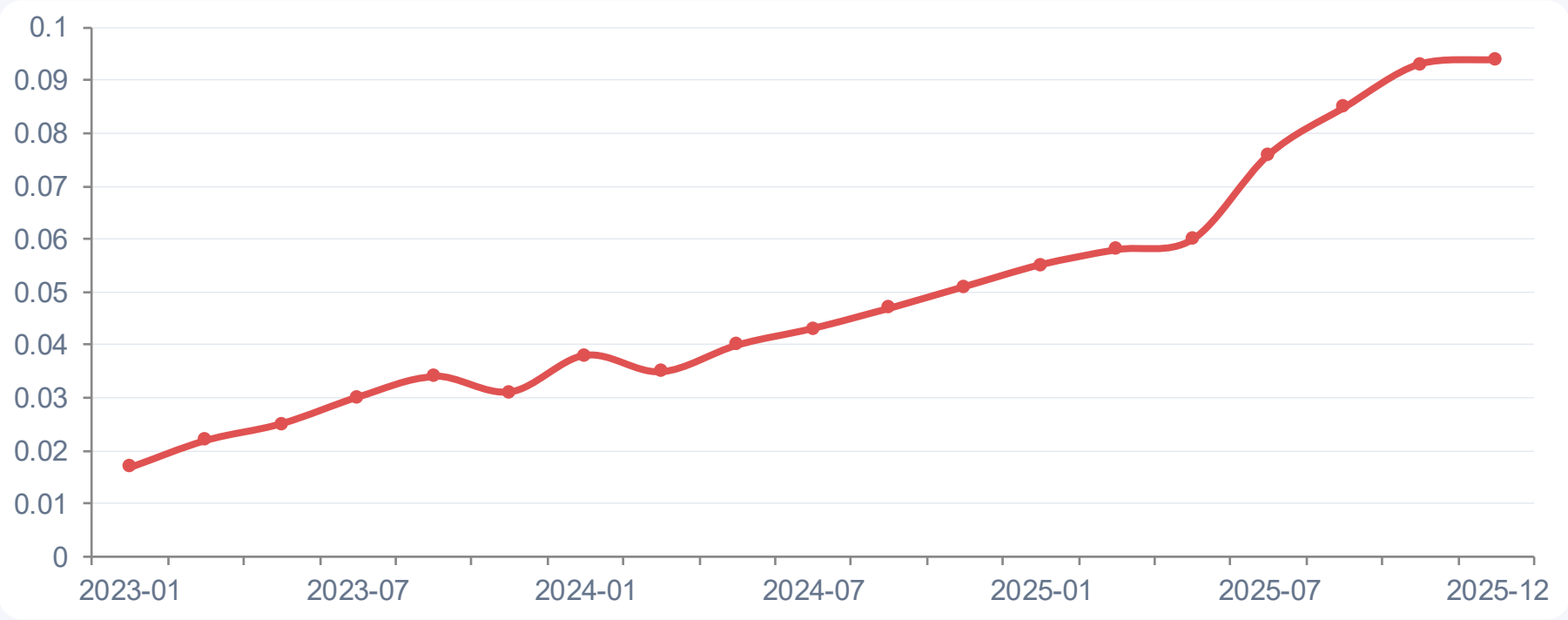
	← Actual A	Actual Non-A →
	11,615 True Negatives	**29** False Positives
	990 False Negatives ⚠	**28** True Positives

⚠ **990 non-A inspections were missed (false negatives) — a major challenge for inspection prioritization.**

Non-A Risk by Establishment Type

All top-risk categories are HIGH RISK establishments. Seat count (61-150) carries the highest non-A rate at 8.6%.

Non-A Rate Trend Over Time (2023–2025)



📈 **Non-A rates have risen from ~2% in early 2023 to ~9.4% by Dec 2025 — a nearly 5× increase.**

Conclusions & Next Steps

✅ Key Findings

- Historical signal exists — prior inspection behavior is predictive of future risk.
- Gradient Boosting (ROC-AUC 0.785) was the best model overall.
- High-risk, larger-capacity restaurants (61-150 seats) carry the highest non-A rates (~8.6%).
- Non-A rates have risen nearly 5× between 2023 and 2025 — a concerning trend.

🔭 Next Steps

- Tune classification threshold — lower it to boost recall and catch more non-A facilities.
- Add geographic features (city, zip) and violation-level detail if available.
- Investigate the post-2024 rise in non-A rates — is it enforcement, reporting, or systemic?
- Deploy as a risk-prioritization tool for inspection scheduling.